METRIC

DOD-C-63980 (AR)
AMENDMENT 9
07 August 1996
SUPERSEDING
AMENDMENT 8
29 November 1994

MILITARY SPECIFICATION

CARTRIDGE, 30MM TARGET PRACTICE, M788

This amendment forms a part of DOD-C-63980 (AR), dated 14 May 1982, and is approved for use by the US Army Armament, Research, Development and Engineering Center, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 2

2.1.2 (Drawings): Delete "11825969 - Test Barrel, 30MM" and substitute "9390627 - Barrel, Test, 30MM (ADEN Barrel)".

Add drawing: "9390748 - Barrel, Test: Gun, M230, 30mm (Progressive w/6.5 degree exit twist)".

2.1.2 (Product and Packaging Drawings): Add "12902718 - Packing and Marking of Ammunition Container, 30mm, M592 for 30mm Cartridges" and "12902719 - Strip, Ammunition, Metallic".

PAGE 3

3.7: Delete in its entirety and substitute the following:

"3.7 Muzzle velocity. The projectile shall have a mean muzzle velocity of 795-815 meters per second (m/s) at 21 degrees C ± 5 degrees C. The standard deviation shall not exceed 15 m/s at temperatures of 21 degrees C ± 5 degrees C and 71 degrees C ± 5 degrees C. The standard deviation shall not exceed 20 m/s at a temperature of 54 degrees C ± 5 degrees C. Following the exposure to the environmental conditions of paragraph 3.9, the mean velocity at 21 degrees C ± 5 degrees C shall not increase or decrease from the mean muzzle velocity before exposure at 21 degrees C ± 5 degrees C by more than 30 m/s."

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PAGE 4

3.8 Delete in its entirety and substitute the following:

"3.8 Case mouth pressure. In the M230 test barrel (Dwg 9390748) the mean peak pressure shall not exceed 335 MPa at 21°C ± 5°C. The mean peak pressure shall not exceed 430 MPa at temperatures over the range from -54°C to 71°C. Following the exposure to environmental conditions of paragraph 3.9, the mean peak pressure shall not increase nor decrease from the mean peak pressure at 21°C ± 5°C by more than 60 MPa."

3.9 Delete the title "Environmental Conditions." and substitute "Environmental exposure."

* 3.9.1: Delete "Method 504.1" and substitute "Method 518".

* 3.9.2: Delete "MIL-STD-331, Method 107" and substitute "MIL-STD-810, Method 509."

3.10: Delete "Drawing 11825969" and substitute "Drawing 9390627".

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4.4.2.4, Minor 201, under Paragraph Reference/Inspection method column: Add "6.8".

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Add Quality Conformance Inspection Sheets:

"4.4.2.6 Packing and marking of ammunition Container, M592 for 30mm Cartridges — Drawing No. 12902718

CRITICAL				
1.	Strip improper (anyone of the forty-four (44) tabs of strip not in groove strip inverted; strip damaged; strip distorted.	100%	3.1.1	Visual
MAJOR				
101.	Cartridge properly mounted	.40%	3.1.1	Visual
102.	Correct number of cartridges	.40%	3.1.1	Visual
103.	Fillers, liners, strip, strap and channels missing	.40%	3.1.1	Visual

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104.	Edge guard channel not properly mounted	.40%	3.1.1	Visual
105.	Contents loose	.40%	3.1.1	Visual

MINOR

201.	Container in proper condition	.65%	3.1.1	Visual
202.	Metallic seal missing	.65%	3.1.1	Visual
203.	Marking improper or illegible	.65%	3.1.1	Visual"

Page 15 should read 15A

PAGE 19

Delete the paragraph "7/ Primer Leaks." in its entirety and substitute the following:

"7/ Primer Leaks. Primer leaks shall be defined as follows:

a. Suspect. A suspect primer leak is defined as the definite evidence of smudging or discoloration around any part of the periphery of the primer cup, button, insulator, or primer pocket. The fired primer must be removed by sectioning the cartridge case and must be disassembled to determine whether actual gas leakage has occurred.

b. Internal minor leak. A suspect primer, which upon disassembly, shows definite evidence of a leakage path between primer components or through a crack or split in any primer component.

c. External minor primer leak. A suspect primer, which upon disassembly, shows definite evidence of a leakage path between the primer cup and primer pocket without evidence of erosion of the pocket.

d. Major leak. A primer leak which shows erosion of any primer component.

e. Critical leak. A primer leak which shows evidence of erosion of the primer pocket or melting of any primer component.

f. A dislodged primer button is not cause for rejection."

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4.5.4 Action Time:

Delete last sentence and substitute the following:

"Any corrected action time of 1.92 milliseconds or less shall be disregarded as instrumentation error."

PAGE 23

4.5.6: Delete in its entirety and substitute the following:

"4.5.6 Test barrel. Test samples for all firing tests shall be fired from a test barrel conforming to Drawing 9390748. (See 6.6 for criteria of barrel replacement.)"

* 4.5.9.1: Delete "Method 514.4" and substitute "Method 518".

* 4.5.9.2: Delete "MIL-STD-331, Method 107" and substitute "MIL-STD-810, Method 509."

4.5.9.3, second sentence: Delete "The test will be conducted with an AAH Magazine.", and substitute "The test will be conducted with an M592 ammo can."

Third sentence: Delete "The vibration...figure 2." and substitute "The vibration...as shown in figures 1 and 2."

PAGE 24

* Add corrected Figure 1, Vibration test curves for equipment installed in helicopters, equipment category C, as included page 6(24).

4.5.13 Accuracy; second sentence:

* Delete "...The target shall be positioned at approximately 25 meters from the gun muzzle. ...", and substitute:

"... The target shall be positioned at 200 feet plus or minus 5 feet from the gun muzzle. ...".

Add the following after the last sentence:

"Alternate procedure at 25 meters +/- 1 meter may be used with

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approved automated electronic accuracy systems (see 6.15)."

PAGE 25

* Add corrected Figure 2, Logarithmic sweep, as included page 7(25).

PAGE 26

Add new paragraph 5.1.1.3 as follows:

"5.1.1.3 Bulk (metallic strip) packing. The Bulk (Metallic Strip) Pack shall consist of ten (10) 30mm Metallic Ammunition Strips (see drawing 12902719) containing 11 cartridges each packed in the M592 shipping and storage container."

Add new paragraph 5.1.2.3 as follows:

"5.1.2.3 Bulk (metallic strip) packing. The interior packing shall be in accordance with drawing number 12902718. The 30mm rounds shall be placed in the Metallic Strips in accordance with the drawing 12910289."

5.2: Delete in its entirety and substitute the following:

"5.2 Marking. Marking for shipment shall be in accordance with the instructions of the contract and shall comply with drawing number 11825893 (for Bulk "Metallic Strip" Packing marking shall comply with drawing number 12902718). In addition, exterior containers shall show the caution marking required by Title 49, Code of Federal Regulations, indicating the appropriate classification of ammunition."

PAGE 24

4.5.13 Accuracy: Change from: "The target shall be positioned at approximately 25 meters from the gun muzzle..." to: "... the target shall be positioned at 200 feet plus or minus 5 feet from the gun muzzle...".

Add the following after the last sentence: "Alternate procedure at 25 meters +/-1 meter may be used with approved automated electronic accuracy system (see 6.15)."

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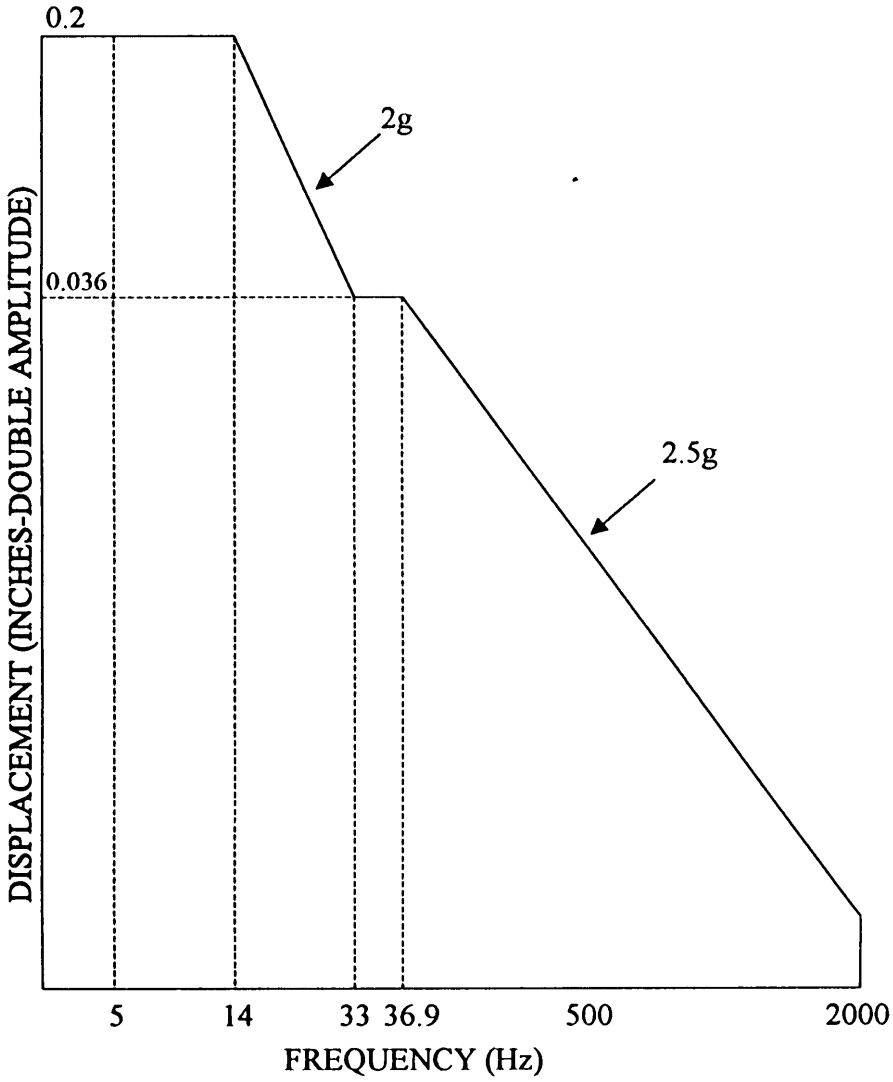


Figure 1. Vibration Test Curves for Equipment Installed in Helicopters, Equipment Category c. (Ref Figure 5.14.2-3 of MIL-STD-810C)

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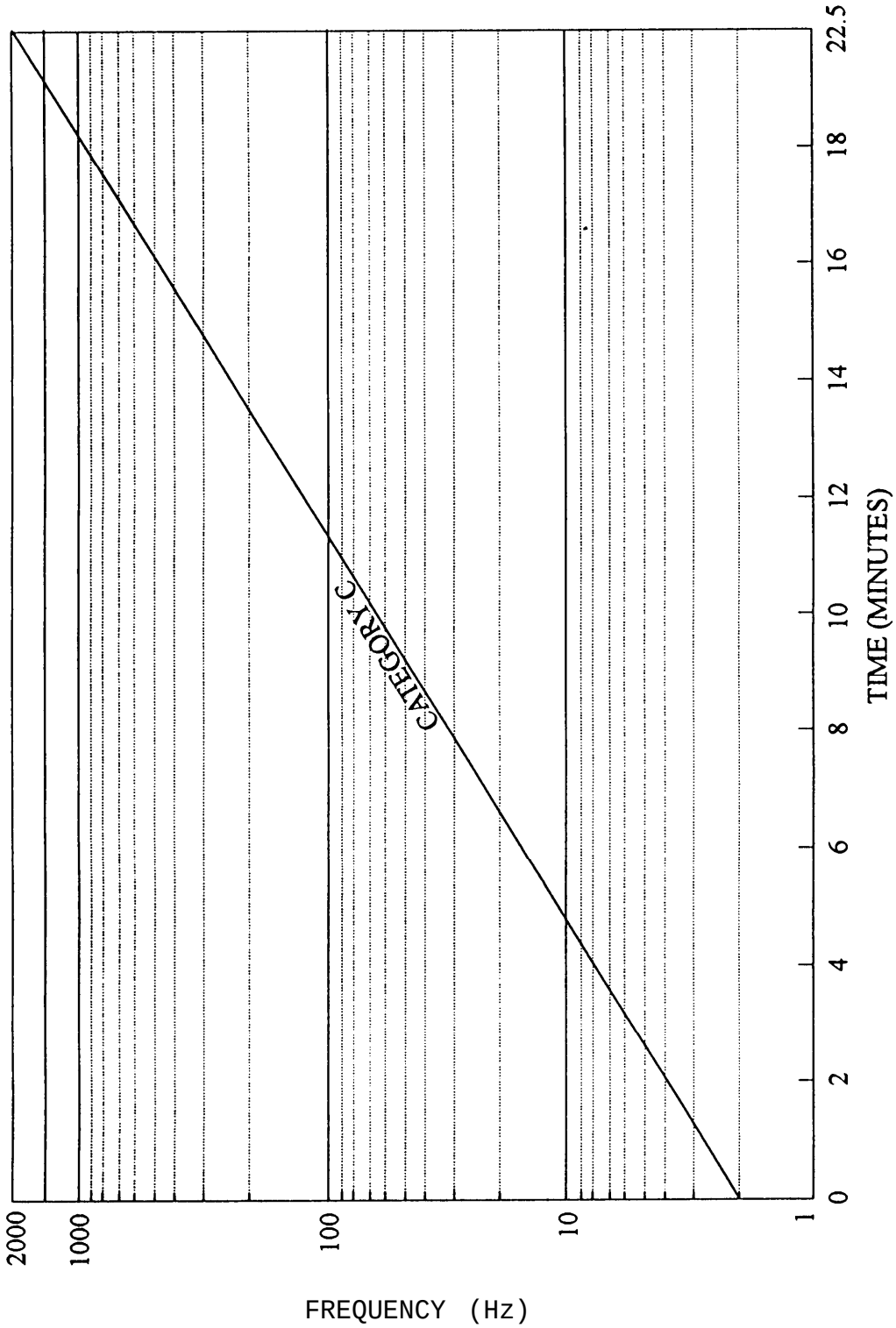


FIGURE 2. Logarithmic Sweep. (Ref. Figure 5.14.2-10 of MIL-STD-810C)

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6.9: Add "d" as follows:

"d. Missing paint on the rotating band is not a defect."

PAGE 29

6.14: Delete in its entirety and substitute the following:

"6.14 Contamination of explosive components. Particular care should be exercised to avoid contamination of primers and propellant by oil or grease. Such contamination may adversely affect performance. Particular care should also be exercised to avoid contamination of primers and propellant by dust and other foreign matter. If contamination by foreign matter or dust is evident, the quantity shall not exceed the requirements set forth in Drawing 11825946."

6.15 Delete "DRDAR-QAF-I" and substitute "AMSTA-AR-QAC-C".

6.16 Delete in its entirety and substitute the following:

"6.16 Retest procedures. Failure of the material sample to comply with the velocity, pressure or action time requirements shall result in a retest. Sample size shall be as indicated in the applicable table."

PAGE 30

6.18: Delete in its entirety.

The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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